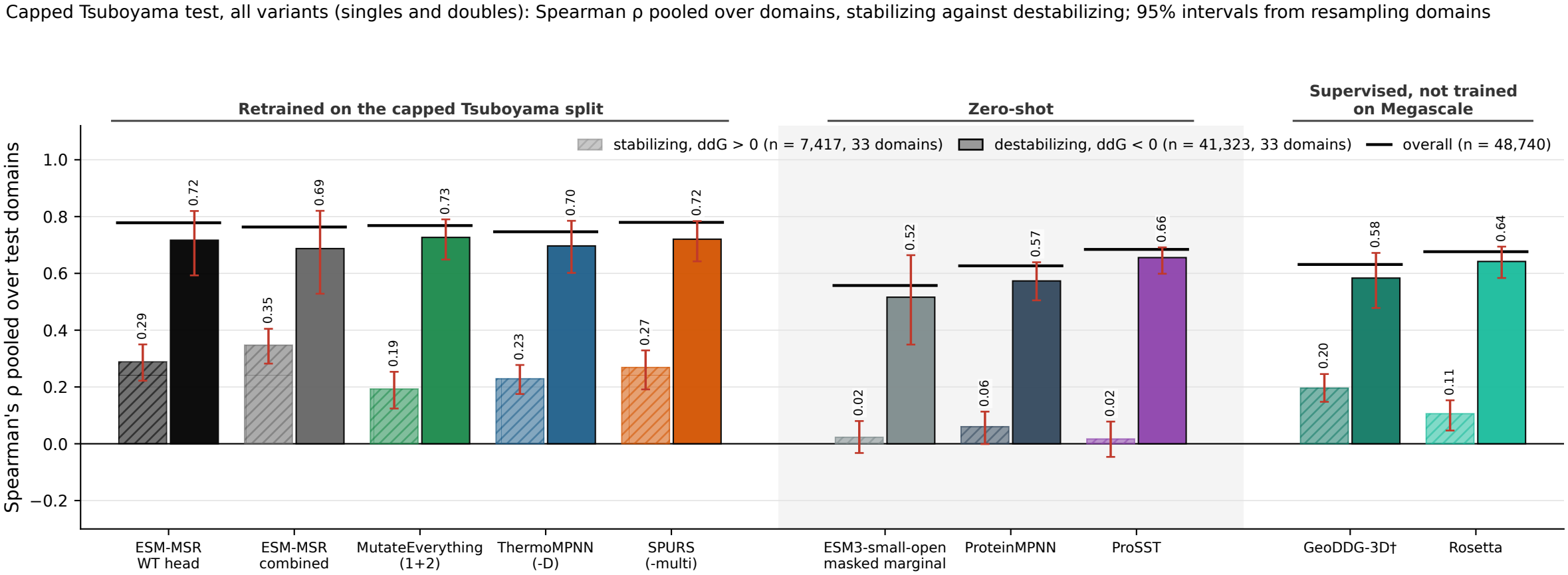

MutateEverything(1+2), ThermoMPNN(-D), SPURS(-multi): singles are scored by the first model, doubles by the second (stage 1 / stage 2, ThermoMPNN / ThermoMPNN-D, SPURS / SPURS multi).
† trained on S8754 (ProThermDB + ThermoMutDB), not on Megascale; chosen over DDGemb by mean per-library singles ρ. ESM3ΔG is not shown: homology to its MGnify fine-tune confounds the comparison.